1. Працююча Kafka і Топіки:

```
(venv) bulga@ndk:/mnt/c/Goit/goit-de-hw-05$ ss -tulpn | grep 9092
tcp   LISTEN 0      50                 *:9092             *:*    users:(("java",pid=118932,fd=170))
(venv) bulga@ndk:/mnt/c/Goit/goit-de-hw-05$ python create_topics.py
Topics created
(venv) bulga@ndk:/mnt/c/Goit/goit-de-hw-05$ ~/kafka/bin/kafka-topics.sh --list --bootstrap-server localhost:9092
building_sensors_dmytro
humidity_alerts_dmytro
temperature_alerts_dmytro
```

2. Генерація даних сенсорів та відправка даних в building_sensors

```
(venv) bulga@ndk:/mnt/c/Goit/goit-de-hw-05$ python -m producer.sensor_producer
Sensor started: 1689
Sending data to: building_sensors_dmytro
/mnt/c/Goit/goit-de-hw-05/producer/sensor_producer.py:28: DeprecationWarning: datetime.datetime.utcnow() is deprecated and scheduled for removal in
 a future version. Use timezone-aware objects to represent datetimes in UTC: datetime.datetime.now(datetime.UTC).
  "timestamp": datetime.utcnow().isoformat(),
{'sensor_id': 1689, 'timestamp': '2026-11-06T04:35:15.190783', 'temperature': 34.21, 'humidity': 73.18}
{'sensor_id': 1689, 'timestamp': '2026-11-06T04:35:20.311089', 'temperature': 28.01, 'humidity': 69.11}
{'sensor_id': 1689, 'timestamp': '2026-11-06T04:35:25.315844', 'temperature': 29.75, 'humidity': 51.65}
{'sensor_id': 1689, 'timestamp': '2026-11-06T04:35:30.318624', 'temperature': 41.16, 'humidity': 44.56}
{'sensor_id': 1689, 'timestamp': '2026-11-06T04:35:35.321205', 'temperature': 43.33, 'humidity': 67.54}
{'sensor_id': 1689, 'timestamp': '2026-11-06T04:35:40.323081', 'temperature': 39.23, 'humidity': 20.23}
{'sensor_id': 1689, 'timestamp': '2026-11-06T04:35:45.329101', 'temperature': 43.78, 'humidity': 18.18}
{'sensor_id': 1689, 'timestamp': '2026-11-06T04:35:50.333250', 'temperature': 35.93, 'humidity': 41.81}
{'sensor_id': 1689, 'timestamp': '2026-11-06T04:35:55.334978', 'temperature': 29.49, 'humidity': 15.77}
{'sensor_id': 1689, 'timestamp': '2026-11-06T04:36:00.337452', 'temperature': 32.01, 'humidity': 61.59}
{'sensor_id': 1689, 'timestamp': '2026-11-06T04:36:05.340962', 'temperature': 39.45, 'humidity': 53.72}
{'sensor_id': 1689, 'timestamp': '2026-11-06T04:36:10.346157', 'temperature': 32.12, 'humidity': 22.37}
{'sensor_id': 1689, 'timestamp': '2026-11-06T04:36:15.348552', 'temperature': 34.62, 'humidity': 28.59}
```

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```
(venv) bulga@ndk:/mnt/c/Goit/goit-de-hw-05$ python -m producer.sensor_producer
Sensor started: 9920
Sending data to: building_sensors_dmytro
/mnt/c/Goit/goit-de-hw-05/producer/sensor_producer.py:28: DeprecationWarning: datetime.datetime.utcnow() is deprecated and scheduled for removal in
 a future version. Use timezone-aware objects to represent datetimes in UTC: datetime.datetime.now(datetime.UTC).
  "timestamp": datetime.utcnow().isoformat(),
{'sensor_id': 9920, 'timestamp': '2026-11-06T04:36:13.212384', 'temperature': 39.99, 'humidity': 37.65}
{'sensor_id': 9920, 'timestamp': '2026-11-06T04:36:18.319949', 'temperature': 40.92, 'humidity': 31.09}
{'sensor_id': 9920, 'timestamp': '2026-11-06T04:36:23.325243', 'temperature': 36.4, 'humidity': 60.56}
{'sensor_id': 9920, 'timestamp': '2026-11-06T04:36:28.327518', 'temperature': 41.83, 'humidity': 55.4}
{'sensor_id': 9920, 'timestamp': '2026-11-06T04:36:33.333102', 'temperature': 37.5, 'humidity': 39.18}
{'sensor_id': 9920, 'timestamp': '2026-11-06T04:36:38.335254', 'temperature': 44.29, 'humidity': 52.87}
{'sensor_id': 9920, 'timestamp': '2026-11-06T04:36:43.338843', 'temperature': 32.2, 'humidity': 45.58}
{'sensor_id': 9920, 'timestamp': '2026-11-06T04:36:48.344434', 'temperature': 26.6, 'humidity': 53.71}
{'sensor_id': 9920, 'timestamp': '2026-11-06T04:36:53.349549', 'temperature': 32.9, 'humidity': 59.23}
{'sensor_id': 9920, 'timestamp': '2026-11-06T04:36:58.351607', 'temperature': 36.31, 'humidity': 41.55}
```

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```
(venv) bulga@ndk:/mnt/c/Goit/goit-de-hw-05$ python -m producer.sensor_producer
Sensor started: 9866
Sending data to: building_sensors_dmytro
/mnt/c/Goit/goit-de-hw-05/producer/sensor_producer.py:28: DeprecationWarning: datetime.datetime.utcnow() is deprecated and scheduled for removal in
 a future version. Use timezone-aware objects to represent datetimes in UTC: datetime.datetime.now(datetime.UTC).
  "timestamp": datetime.utcnow().isoformat(),
{'sensor_id': 9866, 'timestamp': '2026-11-06T04:36:33.911214', 'temperature': 25.84, 'humidity': 34.38}
{'sensor_id': 9866, 'timestamp': '2026-11-06T04:36:39.012191', 'temperature': 36.63, 'humidity': 45.37}
{'sensor_id': 9866, 'timestamp': '2026-11-06T04:36:44.014906', 'temperature': 41.03, 'humidity': 78.88}
{'sensor_id': 9866, 'timestamp': '2026-11-06T04:36:49.016922', 'temperature': 27.71, 'humidity': 72.74}
{'sensor_id': 9866, 'timestamp': '2026-11-06T04:36:54.020282', 'temperature': 28.98, 'humidity': 32.93}
{'sensor_id': 9866, 'timestamp': '2026-11-06T04:36:59.022460', 'temperature': 32.69, 'humidity': 54.34}
{'sensor_id': 9866, 'timestamp': '2026-11-06T04:37:04.024468', 'temperature': 44.74, 'humidity': 63.84}
{'sensor_id': 9866, 'timestamp': '2026-11-06T04:37:09.027384', 'temperature': 26.33, 'humidity': 55.47}
{'sensor_id': 9866, 'timestamp': '2026-11-06T04:37:14.029594', 'temperature': 37.03, 'humidity': 24.21}
{'sensor_id': 9866, 'timestamp': '2026-11-06T04:37:19.032220', 'temperature': 41.51, 'humidity': 57.86}
{'sensor_id': 9866, 'timestamp': '2026-11-06T04:37:24.035461', 'temperature': 31.37, 'humidity': 18.91}
```

3. Отримання даних з `building_sensors`, фільтрація даних
+
4. Відправка фільтрованих даних в `temperature_alerts` та `humidity_alerts`

```
RECEIVED: {'sensor_id': 1416, 'timestamp': '2026-11-06T04:01:10.094501', 'temperature': 37.97, 'humidity': 38.45}

RECEIVED: {'sensor_id': 9727, 'timestamp': '2026-11-06T04:01:15.066005', 'temperature': 44.28, 'humidity': 28.7}
TEMP ALERT GENERATED: {'sensor_id': 9727, 'timestamp': '2026-11-06T04:01:15.066005', 'temperature': 44.28, 'message': 'Temperature exceeded: 44.28
C'}
SENT TO TOPIC: temperature_alerts_dmytro

RECEIVED: {'sensor_id': 1416, 'timestamp': '2026-11-06T04:01:15.095521', 'temperature': 37.91, 'humidity': 49.82}

RECEIVED: {'sensor_id': 9727, 'timestamp': '2026-11-06T04:01:20.067764', 'temperature': 39.52, 'humidity': 61.98}

RECEIVED: {'sensor_id': 1416, 'timestamp': '2026-11-06T04:01:20.096049', 'temperature': 33.15, 'humidity': 55.17}

RECEIVED: {'sensor_id': 9727, 'timestamp': '2026-11-06T04:01:25.068670', 'temperature': 43.92, 'humidity': 44.78}
TEMP ALERT GENERATED: {'sensor_id': 9727, 'timestamp': '2026-11-06T04:01:25.068670', 'temperature': 43.92, 'message': 'Temperature exceeded: 43.92
C'}
SENT TO TOPIC: temperature_alerts_dmytro

RECEIVED: {'sensor_id': 1416, 'timestamp': '2026-11-06T04:01:25.096512', 'temperature': 43.39, 'humidity': 70.86}
TEMP ALERT GENERATED: {'sensor_id': 1416, 'timestamp': '2026-11-06T04:01:25.096512', 'temperature': 43.39, 'message': 'Temperature exceeded: 43.39
C'}
SENT TO TOPIC: temperature_alerts_dmytro
```

5. Отримання даних з `temperature_alerts` і `humidity_alerts` та виведення на екран

```
==============================
TOPIC: humidity_alerts_dmytro
{'sensor_id': 9920, 'timestamp': '2026-11-06T04:45:48.672221', 'humidity': 15.42, 'message': 'Humidity out of range: 15.42%'}
==============================

==============================
TOPIC: temperature_alerts_dmytro
{'sensor_id': 9866, 'timestamp': '2026-11-06T04:45:49.337682', 'temperature': 44.65, 'message': 'Temperature exceeded: 44.65 C'}
==============================

==============================
TOPIC: humidity_alerts_dmytro
{'sensor_id': 9866, 'timestamp': '2026-11-06T04:45:54.340546', 'humidity': 83.22, 'message': 'Humidity out of range: 83.22%'}
==============================

==============================
TOPIC: temperature_alerts_dmytro
{'sensor_id': 9920, 'timestamp': '2026-11-06T04:46:03.679388', 'temperature': 44.83, 'message': 'Temperature exceeded: 44.83 C'}
==============================
```